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$$9.7) \quad a_{\text{obs}}(\vec{r}) = i a_0 \vec{k} \cdot \hat{n}' \frac{e^{ikr}}{2\pi r} \int_{S_A} f_0(r') e^{-i\Delta k \cdot r'} dS'$$

$$\Delta \vec{k} = \vec{k} - \vec{k}_0 \quad \# \quad \vec{k} = k \hat{r}$$

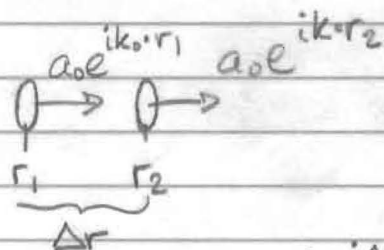
$f_0(r) \equiv$ aperture function $f_0 = 1$ in aperture

$f_0 = 0$ outside aperture

For circular aperture $a(r) = i a_0 R^2 (\vec{k} \cdot \hat{n}') \frac{J_1(\Delta k R)}{\Delta k R} \frac{e^{ikr}}{r}$

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a) for a plane wave $a_0 e^{ik_0 \cdot r_1}$ coming out of aperture 1
plane wave 2 $a_0 e^{ik_0 \cdot r_2}$



then at test point P
we will see
 $a_0 e^{ik_0 \cdot r_1} + a_0 e^{ik_0 \cdot r_2}$

$$\begin{aligned} &= a_0 (e^{ik_0 \cdot r_1} + e^{ik_0 \cdot r_2}) \\ &= a_0 (e^{i\Delta k \cdot \Delta r} + 1) e^{ik_0 \cdot r_2} \\ &= a_0 (e^{i\frac{1}{2}\Delta k \cdot \Delta r} + e^{-i\frac{1}{2}\Delta k \cdot \Delta r}) e^{ik_0 \cdot r_2} e^{i\frac{1}{2}\Delta k \cdot \Delta r} \\ &= 2 \cos(\frac{1}{2}\Delta k \cdot \Delta r) \end{aligned}$$

so then we have $2a_0 \cos(\frac{1}{2}\Delta k \cdot \Delta r)$

to find the intensity take it # square w/ complex conjugate

$$\Rightarrow 4a_0 a_0^* \cos^2(\frac{1}{2}\Delta k \cdot \Delta r)$$

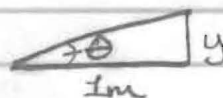
↳ this should also work generally for two slits side by side

$$b) \quad \Theta_1 = 1.22 \frac{\lambda}{D} \Rightarrow \frac{1}{2} \Delta k \cdot \Delta r = 1.22 \frac{\lambda}{D} \quad \Delta k = 1.22$$

$$D = 100 \mu\text{m} \quad \text{distance of } 1\text{m from apertures}$$

$$\lambda = 500 \mu\text{m}$$

$$\Delta r = 150 \mu\text{m}$$



$$\tan \Theta = \frac{y}{1\text{m}}$$

$$\text{Plot } \frac{R^4 J_1^2(kR\Theta) \cdot \cos^2(\frac{1}{2}k\Delta r\Theta)}{r^2 (kR\Theta)^2} \quad k = \frac{2\pi}{\lambda}$$

$$\Rightarrow \frac{(50 \mu\text{m})^4 J_1^2\left(\frac{2\pi}{500 \mu\text{m}} 50 \mu\text{m} \Theta\right) \cdot \cos^2\left(\frac{1}{2} \frac{2\pi}{150 \mu\text{m}} (150 \mu\text{m}) \Theta\right)}{1\text{m}^2 \left(\frac{2\pi}{500 \mu\text{m}} 50 \mu\text{m} \Theta\right)^2}$$

Problem 1b)

In collaboration
w/ Jessica Brinson

$$\text{In}[1]:= c = 3 \cdot 10^8$$

$$\text{Out}[1]= 300\,000\,000$$

$$\text{In}[2]:= \omega = c k$$

$$\text{Out}[2]= 300\,000\,000 k$$

$$\text{In}[3]:= \lambda = 500 \cdot 10^{-9}$$

$$\text{Out}[3]= \frac{1}{2\,000\,000}$$

$$\text{In}[4]:= k = \frac{2\pi}{\lambda}$$

$$\text{Out}[4]= 4\,000\,000 \pi$$

$$\text{In}[5]:= R = 25 \cdot 10^{-6}$$

$$\text{Out}[5]= \frac{1}{40\,000}$$

$$\text{In}[6]:= \Delta r = 150 \cdot 10^{-6}$$

$$\text{Out}[6]= \frac{3}{20\,000}$$

$$\text{In}[7]:= r = 1$$

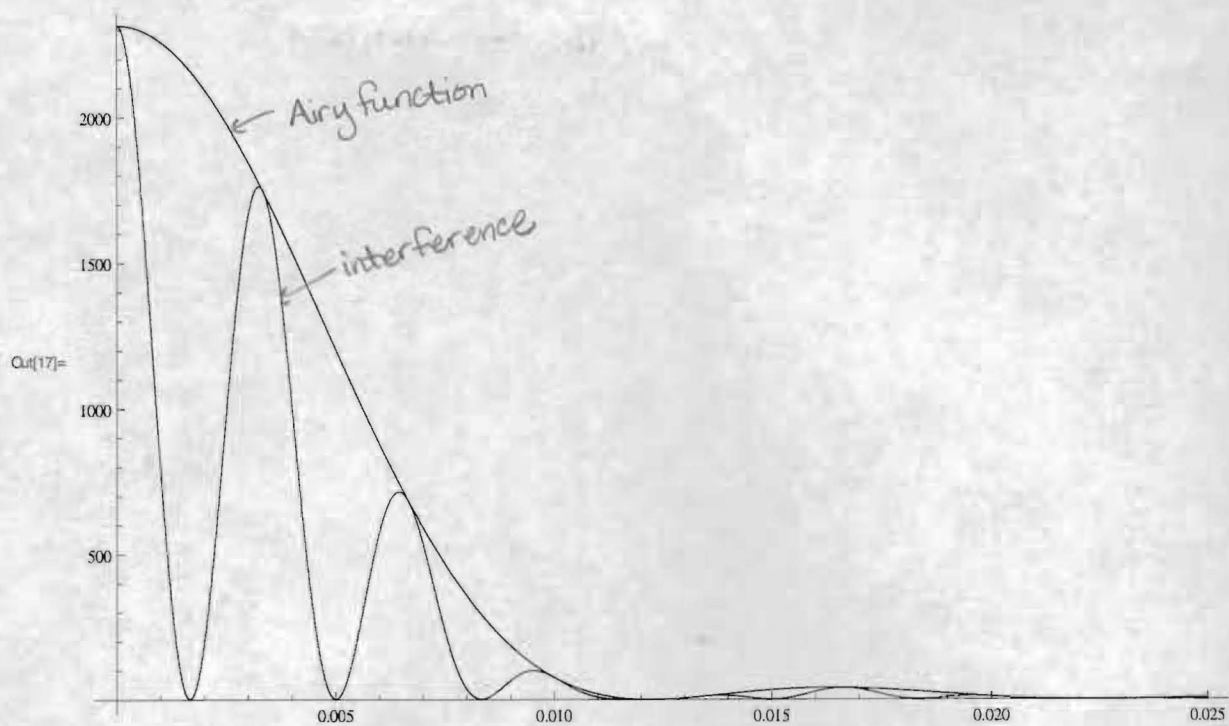
$$\text{Out}[7]= 1$$

$$\text{In}[8]:= \theta[y_] := \text{ArcTan}\left[\frac{y}{r}\right]$$

$$\text{In}[9]:= S[y_] := \frac{\omega k R^4}{2 r^2} \frac{\text{BesselJ}[1, k R \theta[y]]^2}{(k R \theta[y])^2} \left(\cos\left[\frac{1}{2} (k \theta[y] * \Delta r)\right] \right)^2$$

$$\text{In}[10]:= S2[y_] := \frac{\omega k R^4}{2 r^2} \frac{\text{BesselJ}[1, k R \theta[y]]^2}{(k R \theta[y])^2}$$

```
In[17]:= Plot[{S[y], S2[y]}, {y, 0, 1000 R}]
```



Where we have plotted the Airy function envelope in pink and the diffraction pattern observed in blue

Problem 2c)

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In[15]:= Sum[x^n, {n, N}]
```

$$\text{Out[15]} = \frac{x(-1 + x^N)}{-1 + x}$$

10/10

$$2) a) a_0 = i a_0 \vec{k} \cdot \hat{n}' e^{i k r} \int f_n(r') \sum_{n=1}^N e^{-i \Delta k \cdot (r' - r_n)} dS'$$

$$\tilde{f}(\Delta k) = \frac{1}{2\pi} \int dS' f_n(r') e^{-i \Delta k r'}$$

$$\therefore \text{we get } a_0 = i a_0 \vec{k} \cdot \hat{n}' e^{i k r} \underbrace{\frac{1}{r} \int dS' f_n(r') e^{-i \Delta k r'}}_{\tilde{f}(\Delta k)} \sum_{n=1}^N e^{i \Delta k \cdot r_n}$$

$$\Rightarrow a_0 = i a_0 \vec{k} \cdot \hat{n}' e^{i k r} \tilde{f}(\Delta k) \sum_{n=1}^N e^{i \Delta k \cdot r_n}$$

$$2) b) \hat{f}(\Delta k) = \frac{1}{2\pi} \int_{S'} e^{i\Delta k \cdot r'} f(r') dS'$$

$$= \frac{1}{2\pi} \int_{S'} e^{i\Delta k_x x} e^{i\Delta k_y y} dx dy = \frac{1}{2\pi} \int_0^l \int_0^w e^{i\Delta k_x x} e^{i\Delta k_y y} dx dy$$

$$= \frac{1}{2\pi} \left. e^{i\Delta k_x x} \right|_0^l \left. e^{i\Delta k_y y} \right|_0^w$$

$$= \frac{1}{2\pi} \frac{(e^{i\Delta k_x l} - 1)}{i\Delta k_x} \frac{(e^{i\Delta k_y w} - 1)}{i\Delta k_y}$$

$$(e^{i\Delta k_x l} - 1) = \left(e^{\frac{i}{2}\Delta k_x l} - e^{-\frac{i}{2}\Delta k_x l} \right) e^{-\frac{i}{2}\Delta k_x l}$$

$$= 2i \sin\left(\frac{1}{2}\Delta k_x l\right)$$

$$\Rightarrow \frac{4}{2\pi} \frac{\sin\left(\frac{1}{2}\Delta k_x l\right) \sin\left(\frac{1}{2}\Delta k_y w\right)}{\Delta k_x \Delta k_y}$$

$$= \frac{2}{\pi} \frac{\sin\left(\frac{1}{2}\Delta k_x l\right) \sin\left(\frac{1}{2}\Delta k_y w\right)}{\Delta k_x \Delta k_y}$$

$$c) \text{ By mathematica } \sum_{n=1}^N x^n = \frac{x(x^N - 1)}{x - 1}$$

→ see earlier
mathematica
print out

$$\text{Show that } \sum_{n=1}^N x^n = e^{\frac{i}{2}(N+1)\phi} \frac{\sin\left(\frac{1}{2}N\phi\right)}{\sin\left(\frac{1}{2}\phi\right)}$$

$$\sum_{n=1}^N x^n = \frac{x(x^N - 1)}{x - 1} \quad \text{for } x = e^{i\phi}$$

$$\Rightarrow \frac{e^{i\phi}(e^{iN\phi} - 1)}{e^{i\phi} - 1}$$

$$e^{i\phi} - 1 = (e^{i\phi/2} - e^{-i\phi/2})e^{i\phi/2} = 2i\sin(\frac{1}{2}\phi)e^{i\phi/2}$$

$$= \frac{e^{i\phi}(e^{iN\phi} - 1)}{2i\sin(\frac{1}{2}\phi)e^{i\phi/2}}$$

$$(e^{iN\phi} - 1) = 2i\sin(\frac{1}{2}N\phi)e^{iN\phi/2}$$

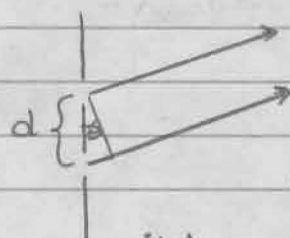
$$= e^{i\phi} \frac{2i\sin(\frac{1}{2}N\phi)e^{iN\phi/2}}{2i\sin(\frac{1}{2}\phi)} e^{-i\phi/2}$$

$$= e^{i\phi/2} e^{iN\phi/2} \frac{\sin(\frac{1}{2}N\phi)}{\sin(\frac{1}{2}\phi)} = e^{i\frac{(N+1)\phi}{2}} \frac{\sin(\frac{1}{2}N\phi)}{\sin(\frac{1}{2}\phi)} \quad \checkmark$$

So show for N apertures $\phi = \Delta k_{\perp} d$

$$\text{From } \sum_{n=1}^N e^{i\vec{k} \cdot \vec{r}_n} = e^{i(\frac{1}{2}N+1)\Delta k_{\perp} d} \frac{\sin(\frac{1}{2}N\Delta k_{\perp} d)}{\sin(\frac{1}{2}\Delta k_{\perp} d)}$$

$\Delta k \cdot r_n$ which we know by the geometry of the problem the path difference is



$$\sin\theta = \frac{\Delta x}{d} \Rightarrow \Delta x = d\sin\theta$$

For small θ we get

$$\text{then for } e^{ik\Delta x} = e^{ikd\theta}$$

by the Fraunhofer approx. $\Delta k = k\theta$

$\Rightarrow$ our phase factor $i\Delta k_{\perp} d \because \phi = \Delta k_{\perp} d$ we get e

$$\sum_{n=1}^N e^{i\vec{k} \cdot \vec{r}_n} = e^{i(\frac{1}{2}N+1)\Delta k_{\perp} d} \frac{\sin(\frac{1}{2}N\Delta k_{\perp} d)}{\sin(\frac{1}{2}\Delta k_{\perp} d)} \quad \checkmark$$

2)d) Show that the peaks occur at $\Delta k_{wd} = m2\pi$

$$\frac{\sin(\frac{1}{2}N\Delta k_{wd})}{\sin(\frac{1}{2}\Delta k_{wd})}$$

where $\sin(\frac{1}{2}\Delta k_{wd}) = 0, \pi, 2\pi, \dots \Rightarrow m\pi$

$$\frac{1}{2}\Delta k_{wd} = m\pi \Rightarrow \Delta k_{wd} = 2m\pi$$

2) e) as we showed in 2) d) we have maxima every $2m\pi$
 & we want to find the minima about these maxima

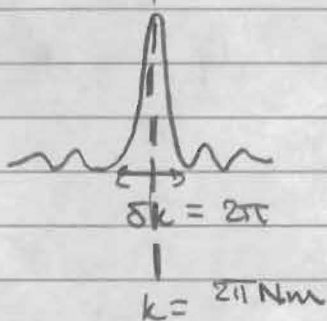
$$\Rightarrow \frac{\sin(Nx)}{\sin(x)} = 0 \Rightarrow x = \frac{m\pi}{N} \quad \& \text{ we want the first minima} \Rightarrow m=1$$

$$\frac{1}{2}\Delta kd = \frac{m\pi}{N} \Rightarrow \Delta kd = \frac{2\pi}{N} \quad \therefore \text{these minima occur when}$$

$$\Delta kd = 2m\pi \pm \frac{2\pi}{N} \Rightarrow N\Delta kd = (Nm \pm 1)2\pi$$

so then for the resolution we can resolve

$$\frac{Nm(2\pi)}{k} \pm \frac{2\pi}{\delta k} \Rightarrow \frac{\delta k}{k} = \frac{2\pi}{Nm(2\pi)} = \frac{1}{Nm} \quad \checkmark$$



$$3) \text{ 9.175) } a_n(\vec{r}) = i a_0 \vec{k}_0 \cdot \hat{n}' \frac{e^{ikr}}{2\pi r} \int_{S_a} e^{-i\Delta\vec{k} \cdot \vec{r}'} dS'$$

normal incidence $\Rightarrow \vec{k}_0 \cdot \hat{n}' = \vec{k}_0 \cdot \hat{n}' \Rightarrow$ we get $-k$

plane aperture $\Rightarrow dS' = dx'dy'$

$r \rightarrow L$

so now we have $a_n(\vec{r}) = -i a_0 k \frac{e^{ikr}}{2\pi L} \int_{S_a} e^{-i\Delta\vec{k} \cdot \vec{r}'} dx'dy'$

$\rightarrow$ pull $e^{ik \cdot r}$ into the integral to get

$$e^{ik \cdot r - i\Delta\vec{k} \cdot \vec{r}'} \quad \Delta\vec{k} \cdot \vec{r}' = (\vec{k} - \vec{k}_0) \cdot \vec{r}' \quad \text{where } \vec{k}_0 \perp \vec{r}' \Rightarrow \vec{k}_0 \cdot \vec{r}' = 0$$

$$= e^{ik \cdot r - ik \cdot r'} = e^{ik \cdot \Delta r} \quad \text{where in 9.185 we get}$$

$$\Delta r \approx L + \frac{r''^2}{2L} + \frac{r'^2}{2L} - \frac{r'' \cdot r'}{L} \quad \text{0 b/c its really small square}$$

then the exponential gives $e^{ikL} e^{\frac{ik}{2L}(x^2+y^2)} e^{-\frac{ik}{L}(xx'+yy')}$

$$\therefore a_n(\vec{r}) = -i a_0 k \frac{e^{ikL}}{2\pi L} e^{\frac{ik}{2L}(x^2+y^2)} \int_{S_a} e^{-\frac{ik}{L}(xx'+yy')} dx'dy'$$

$$9.210 \quad a(x,y) = -ia_0 k \frac{e^{ikL}}{2\pi L} \int_{S_a} dx' dy' e^{\frac{ik(x'-x)^2}{2L}} e^{i(\frac{k}{2L})(y'-y)}$$

$$(x'-x)^2 = x'^2 + x^2 - 2x'x \quad x'^2 \text{ \& } y'^2 \text{ are really small}$$

$$(y'-y)^2 = y'^2 + y^2 - 2y'y$$

$$a(x,y) = -ia_0 k \frac{e^{ikL}}{2\pi L} e^{\frac{ik(x^2+y^2)}{2L}} \int_{S_a} dx' dy' e^{-\frac{ik}{L}(xx' + yy')}$$

- circular aperture
 a) R @ normal incidence by plane wave of amplitude a_0
 wave vector k on axis

$$a^*a = 4a_0^2 \sin^2\left(\frac{kR^2}{4L}\right) \xrightarrow{L \rightarrow \infty} a_0^2 a_0 \left(\frac{k^2 R^4}{4L^2}\right)$$

$$a(x,y) = \frac{-ia_0 k L}{2\pi} \int_{S_a} dx' dy' \frac{e^{ik|\vec{r}' - \vec{r}|}}{|\vec{r}' - \vec{r}|^2}$$

$$|\vec{r}' - \vec{r}| = L^2 + (x' - x)^2 + (y' - y)^2 \quad x=0 \quad y=0$$

$$= L^2 + x'^2 + y'^2 \approx L + \frac{1}{2L}x'^2 + \frac{1}{2L}y'^2$$

$$\frac{-ia_0 k L}{2\pi} \int_{S_a} dx' dy' \frac{e^{ikL + i\frac{k}{2L}(x'^2 + y'^2)}}{L^2 - x'^2 - y'^2}$$

$$= \frac{-ia_0 k L e^{ikL}}{2\pi L^2} \int_0^R r e^{\frac{ikr^2}{2L}} dr d\theta = \frac{-ia_0 k e^{ikL}}{L} \int_0^R r e^{\frac{ikr^2}{2L}} dr$$

$$= +a_0 k \left(\frac{k}{k}\right) e^{ikL} e^{\frac{ikr^2}{2L}} \Big|_0^R = a_0 e^{ikL} \left[e^{\frac{ikR^2}{2L}} - 1 \right]$$

$$= a_0 e^{ikL} \left(e^{\frac{ikR^2}{4L}} - e^{-\frac{ikR^2}{4L}} \right) e^{\frac{ikR^2}{4L}}$$

$$= 2ie^{ikL} e^{\frac{ikR^2}{4L}} \sin\left(\frac{kR^2}{4L}\right)$$

$$\Rightarrow aa^* = 4a_0^2 \sin^2\left(\frac{kR^2}{4L}\right)$$

by small angle approx as $L \rightarrow \infty$ we get

$$4a_0^2 \left(\frac{k^2 R^4}{16L^2}\right) = \frac{a_0^2 k^2 R^4}{4L^2}$$

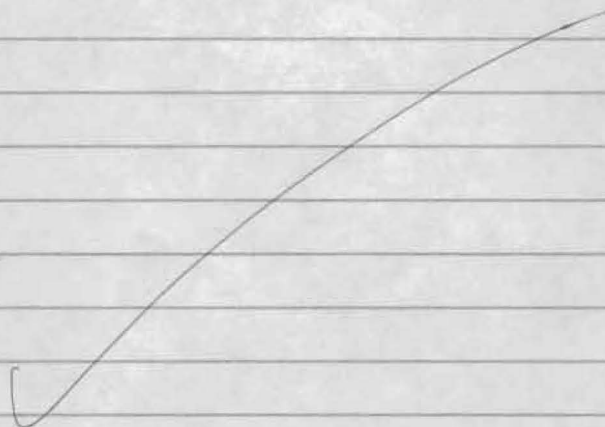
$$4)b) \quad a_0(\vec{r}) = a_0 e^{ikL} \quad a_a = a_0 e^{ikL} [e^{ikR/2L} - 1]$$

$a_s = a_0 - a_a$ by Babinet's principle

$$a_s a_s^* = a_0 e^{ikL} [1 - e^{ikR/2L} + 1] a_0^* e^{-ikL} [1 - e^{-ikR/2L} + 1]$$

$$= a_0 a_0^*$$

$$\therefore a a^* = a_0 a_0^*$$



a) Show that $a^*a = 4N^2 a_0^* a_0$

we showed in 9.10 that for a circular aperture that $a_0 = a_0 2i e^{ikL} \sin\left(\frac{kR^2}{4L}\right)$

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and so for the zone plate we wish to maximize the amplitude so if we maximize $\sin\left(\frac{kR^2}{4L}\right) = 1$ for a particular value of $R, k, \& L$

we get $a = a_0 2i e^{ikL}$ & summing over all these maximized circles all of the same amplitude we get

$a_{\text{total}} = 2a_0 i N e^{ikL}$ & thus the intensity is

$$a_{\text{total}}^* = 4a_0 a_0^* N^2$$

b) for a zone plate of radius R show that $N \approx \frac{R^2}{2L\lambda}$

again we want $\sin^2\left(\frac{kR^2}{4L}\right)$ to be a max

$$\Rightarrow \frac{kR^2}{4L} = \frac{(2n+1)\pi}{2} \Rightarrow \frac{2\pi R^2}{4\lambda L} = \frac{(2n+1)\pi}{2}$$

$$\Rightarrow \frac{\pi R^2}{\lambda L} = (2n+1)\pi \quad \text{so for large } R \text{ we'll have a lot of } n$$

$$\Rightarrow n \approx \frac{R^2}{2\lambda L}$$

$$c) \quad a(\vec{r}) = \frac{-ik}{2\pi r} e^{ik[L + (n-1)t_0 + (r'^2/2L)]} \int_{S_a} a_i(\vec{r}') e^{-i\vec{k} \cdot \vec{r}'} dS'$$

if we are far away $r \rightarrow L$ as in 9.9

$$= \frac{-ik}{2\pi L} e^{i\phi} \int_{S_a} a_i(\vec{r}') e^{-i\vec{k} \cdot \vec{r}'} dS' \quad a_i(\vec{r}) = e^{i\vec{k} \cdot \vec{r}} a_0$$

$$\Rightarrow \frac{-ika_0}{2\pi L} e^{i\phi} \int_{S_a} e^{i\vec{k} \cdot \vec{r}'} dS' \quad \text{which is the same integral we evaluated in \#9.10}$$

$$\Rightarrow aa^* = 4a_0^* a_0 \sin^2\left(\frac{kR^2}{4L}\right) \text{ as } L \rightarrow \infty$$

$$\text{we get } a_0^* a_0 \frac{k^2 R^4}{4L^2} = a_0^* a_0 \frac{4\pi^2 R^4}{\lambda^2 L^2} = a_0^* a_0 \frac{\pi^2 R^4}{\lambda^2 L^2}$$